Software Requirements Specification

PDS Trust — Transparent, Inclusive and Verifiable Public Distribution System

Version 1.0 | 27 October 2026 | Status: Proposed MVP specification (not an implementation claim)

1. Purpose and scope

Define a demonstrable Laravel prototype for transparent ration stock allocation, distribution, beneficiary confirmation, discrepancy review, and grievance resolution. This is a simulated demonstration system; it is not an authorised government PDS integration or an official entitlement system.

2. Stakeholders and access

- Beneficiary: view own simulated entitlements and transaction history, confirm/dispute recorded receipt, submit and track grievances.
- Fair Price Shop (FPS) operator: view assigned stock, acknowledge receipt, record distribution, see discrepancies and respond with evidence.
- Government official: create stock allocations, inspect reconciliation alerts, review cases, resolve grievances, view audit events.
- System administrator (seeded/development only): provision demo accounts and configure test data; cannot silently alter audit events.

3. Assumptions and boundaries

- Only synthetic beneficiaries, shops, phone numbers and transactions are used in the competition demo.
- No Aadhaar, biometric verification, live government database, or actual ration delivery is required.
- SMS is simulated through an internal message outbox and inbound reply form; no real SMS delivery is claimed.
- One demo region, one commodity or a small predefined commodity list, and a small set of FPS shops are sufficient.
- Beneficiary confirmation is a signal for review, not conclusive proof of misconduct.

4. Functional requirements

FR-01 — Authentication and role-based authorisation

Only authenticated users can access their assigned role actions and records.

FR-02 — Beneficiary entitlement view

Display assigned monthly commodity and quantity, clearly marked demo data.

FR-03 — Stock allocation

Officials create allocations to shops with commodity, quantity, date and reference.

FR-04 — Stock receipt

Shop operators acknowledge actual quantity received; differences create a discrepancy.

FR-05 — Inventory ledger

Every stock movement is recorded; current balance is derived from ledger entries.

FR-06 — Distribution entry

Operators record a beneficiary transaction; prevent quantity exceeding available balance or entitlement in the configured demo period.

FR-07 — Transaction history

Beneficiaries see only their own transactions and confirmation status.

FR-08 — SMS outbox simulator

Recording distribution creates a message with transaction ID, quantity and reply instructions; messages remain visible in a demo outbox.

FR-09 — Inbound reply simulator

Authorised demo action submits YES / LESS / NOT_RECEIVED for a transaction using a one-time reply token; duplicate replies are rejected or tracked.

FR-10 — Discrepancy engine

Automatically flag allocation-versus-receipt mismatches and beneficiary disputes; avoid classifying a missing reply as fraud.

FR-11 — Case review

Officials see flagged cases, status, evidence, timestamps and assigned resolution notes.

FR-12 — Grievance management

Beneficiaries submit complaints; officials update status and add response; beneficiary sees outcome.

FR-13 — Audit logging

Record who performed stock, distribution, verification and resolution changes, with timestamp and prior/new values where applicable.

FR-14 — Summary dashboard

Officials view allocations, distribution totals, open discrepancies and pending grievances.

FR-15 — Demo data and tests

Seed sample users, shops, allocations, successful and disputed transactions; provide automated tests for core flows.

FR-16 — Dual-Channel Beneficiary Access

The system shall allow beneficiaries to access ration distribution verification services through either an authenticated web portal or an SMS-based interaction channel. Both channels shall use the same backend verification and discrepancy management workflow. SMS interactions shall not require web login and shall be associated with the appropriate beneficiary and transaction using a registered phone number and transaction-specific verification mechanism.

FR-17 — SMS Transaction Identification

The system shall generate a unique transaction reference for every ration distribution and associate incoming SMS responses with the corresponding transaction. Invalid, duplicate, or expired responses shall be handled safely without modifying previously verified records.

FR-18 — Cross-Channel Consistency

The system shall maintain consistent transaction verification statuses across the web portal and SMS simulation. A verification or dispute submitted through either channel shall be reflected in the other channel and in the government official's dashboard.

FR-19 — Unconfirmed Transaction Handling

The system shall distinguish between confirmed, disputed, and unconfirmed ration distributions. Failure to respond to an SMS shall not be treated as confirmation of receipt.

FR-20 — Accessible Notifications

The system shall support configurable SMS message templates in English and Hindi, including transaction notifications, verification instructions, and complaint reference updates.

5. Core workflow

- Stock Allocation: A government official allocates ration stock to a registered Fair Price Shop (FPS).
- Stock Receipt: The FPS owner records the quantity received. The system compares the allocated and received quantities and flags discrepancies.
- Ration Distribution: The FPS owner records a distribution transaction against an eligible beneficiary, specifying the commodity, quantity, and date. The system generates a unique transaction reference and updates the inventory ledger.
- Beneficiary Notification: The system generates a transaction notification containing the recorded distribution details and verification instructions. For beneficiaries enrolled in SMS notifications, the prototype delivers this through the SMS simulator.
- Dual-Channel Verification: The beneficiary can verify the transaction through either:
    - Web Portal: Log in to view transaction history and confirm or dispute the recorded distribution.
    - SMS Channel: Reply using the transaction-specific reference and supported response options without requiring web login. The competition prototype simulates SMS delivery and replies.
- Verification Status Management: The system maintains a consistent verification status across both channels:
    - Pending: No valid beneficiary response has been received.
    - Confirmed: The beneficiary confirms receiving the recorded ration.
    - Disputed: The beneficiary reports a discrepancy.
- Duplicate, invalid, and expired SMS responses are rejected or handled safely. A lack of response must never be interpreted as confirmation.
- Discrepancy Detection: The system automatically identifies stock mismatches and disputed transactions, generating discrepancy cases linked to the relevant records.

- Government Review: Authorised officials review flagged discrepancies, inspect transaction and inventory records, and record their findings and resolution.
- Grievance Management: Beneficiaries can submit grievances through the web portal. SMS-supported dispute responses create trackable cases, and relevant status notifications are generated through the SMS simulator.
- Audit Logging: The system records critical activities, including allocations, stock receipts, distributions, beneficiary responses, discrepancy creation, and official resolutions, in an application-level append-only audit log.

6. Proposed Data Model

The system shall maintain the following core entities:

Entity	Purpose
Users	Stores authenticated user accounts, roles, and account status for beneficiaries, FPS owners, and government officials.
Beneficiary Profiles	Stores beneficiary information, masked ration card identifiers, registered mobile numbers, preferred language, and SMS consent preferences.
Fair Price Shops	Stores registered FPS details, location, assigned owner, and operational status.
Commodities	Defines ration commodities and their measurement units.
Entitlements	Defines the quantity of each commodity allocated to eligible beneficiaries for a specified distribution period.
Stock Allocations	Records commodity quantities allocated by government officials to Fair Price Shops.
Stock Receipts	Records quantities actually received by Fair Price Shops against allocations.
Inventory Ledger	Maintains an append-only history of stock movements, including receipts, distributions, and authorized adjustments.
Distribution Transactions	Records ration distributions, beneficiary references, commodity quantities, transaction timestamps, unique transaction references, and current verification status.
Verification Responses	Stores beneficiary confirmations or disputes, response channel (web/SMS), timestamps, transaction references, and response validity.
SMS Messages	Stores simulated outgoing and incoming SMS messages, delivery status, associated transactions, and processing results.
Discrepancy Cases	Tracks detected stock mismatches, beneficiary disputes, investigation status, and official resolution.
Grievances	Stores beneficiary complaints, descriptions, status, and resolution history.
Audit Logs	Records critical system actions, actors, timestamps, and relevant entity references for accountability.

Data Integrity Requirements

- Every distribution transaction shall have a unique transaction reference.
- Verification responses shall be associated with an existing distribution transaction and recorded with their source channel.
- Invalid, expired, or duplicate SMS responses shall not overwrite valid verification records.
- Verification statuses shall remain consistent across the web portal, SMS simulator, and official dashboard.

- The system shall preserve historical verification responses, including subsequent disputes.
- Inventory balances shall be derived from the inventory ledger rather than modified directly.
- Discrepancy cases shall reference the relevant allocation, receipt, distribution, or verification records.
- Sensitive beneficiary information shall be protected through role-based access controls and appropriate data minimisation.
- Audit logs shall be append-only for ordinary application users.
- SMS message records shall distinguish simulated messages from actual telecom delivery.

7. Non-functional requirements

- NFR-01 Security: Laravel validation, CSRF protection, hashed passwords, authorisation policies, server-side checks and rate limiting on reply endpoints.
- NFR-02 Privacy: synthetic data for demo; mask identifiers; do not expose phone numbers or beneficiary records across roles.
- NFR-03 Integrity: database transactions for distribution and stock deduction; prevent negative balances and duplicate confirmation processing.
- NFR-04 Accessibility: mobile-responsive screens, clear Hindi/English-ready copy, readable text and keyboard-accessible forms.
- NFR-05 Performance target: typical seeded-data pages respond within 3 seconds in the demo environment (to be measured, not assumed).
- NFR-06 Reliability: meaningful errors and retry-safe processing; no assumption that undelivered or unanswered SMS confirms receipt.
- NFR-07 Reproducibility: documented installation, migrations, seeders, environment variable example, license and automated test instructions.
- NFR-08 Data Privacy: The system shall restrict access to personal beneficiary information according to user roles, minimise sensitive information included in SMS notifications, and avoid exposing complete ration card numbers in publicly accessible interfaces.
- NFR-09 Audit-ability: The system shall maintain an immutable application-level audit history of critical operations, including stock allocation, distribution recording, beneficiary verification, discrepancy creation, and case resolution.

8. Acceptance tests

- AT-01 Beneficiary cannot view another beneficiary’s transactions (403/404).
- AT-02 Shop cannot distribute more stock than available; no stock ledger mutation on failure.
- AT-03 Receipt of 950 kg against 1,000 kg allocation creates one discrepancy case.
- AT-04 A distribution creates exactly one simulated outgoing message.
- AT-05 A valid LESS reply creates or updates one dispute case; replay does not create duplicates.

- AT-06 A missing SMS reply leaves transaction unconfirmed, not fraud-confirmed.
- AT-07 Official resolves a case with a note; audit event records actor and timestamp.
- AT-08 Beneficiary can see their own grievance status but cannot modify official resolution.

9. Future development (explicitly excluded from MVP)

- IVR calls in Hindi and regional languages for voice-based transaction verification and complaints.
- Real SMS provider integration, opt-in consent management and delivery receipts.
- USSD menus for feature phones, offline-first shop synchronisation, anomaly models and government APIs.
- Accessibility evaluation with beneficiaries and safeguards for shared phones, literacy and device access.

10. Risks and mitigations

- Risk: short competition deadline. Mitigation: finish core scenario and automated tests before adding polish.
- Risk: simulated SMS mistaken for live service. Mitigation: label every simulator screen and video demonstration.
- Risk: disputed transaction treated as evidence of fraud. Mitigation: use neutral discrepancy labels and human review.
- Risk: unauthorised access or personal data exposure. Mitigation: use only synthetic data and enforce authorisation.

11. Delivery and traceability

Repository should include Laravel source, README, .env.example, architecture/SDD, schema diagram, seeders, tests, LICENSE and sample screenshots. Mark each requirement as implemented, partially implemented or planned in the final README; this SRS alone is not evidence of implementation.